

AKSHAY MANIYAMPARA

Data Scientist • Machine Learning Engineer • MLOps

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PROFESSIONAL SUMMARY

Machine Learning and Deep Learning practitioner driven by a genuine interest in understanding model mechanics and the mathematics behind algorithms. Applies that foundation to design and ship data-driven AI solutions that address real-world business problems.

EDUCATION

MS Data Science & Spatial Analysis | MS Technical Systems (AI/ML in Space Science) 2026 – 2028

Symbiosis International University & RUDN University, Moscow

Bachelor of Science in Data Science — University of Mumbai 2026

SIES College of Arts, Science and Commerce, Mumbai

EXPERIENCE

Mgrid Technologies PVT LTD • Backend Development Intern • Remote Nov 2025 – Jan 2026

- Built OCR and Named Entity Recognition (NER) pipeline using Python, Google Vision API, and GLiNER Bio-Med to extract medication names, dosages, and schedules from prescription images.
- Deployed containerized backend on Google Cloud Run with automated CI/CD via GitHub Actions for real-time prescription data processing.
- Integrated Twilio API to automate WhatsApp medication reminders for patient adherence tracking.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript

ML / AI Frameworks: PyTorch, TensorFlow, Scikit-learn, XGBoost, NumPy, Pandas

Data Visualization: Power BI, Streamlit, Plotly, Matplotlib, Seaborn

Databases: PostgreSQL, MySQL, MongoDB, BigQuery, NeonDB, Qdrant

Cloud & Deployment: Vercel, AWS, GCP, Docker, GitHub Actions, Hugging Face Spaces, Flask, FastAPI, React, Git

PROJECTS

Vector Auditor — Agentic Document Q&A | Python, FastAPI, Qdrant, Docker, Hugging Face Spaces 2026

- Developed a Retrieval-Augmented Generation (RAG) pipeline with semantic search and cross-encoder reranking for multi-document Q&A over PDFs and DOCX files, delivering page-level citations using SentenceTransformers MiniLM and BAAI/bge-reranker-v2-m3.
- Deployed containerized FastAPI backend on Hugging Face Spaces with JWT and GitHub OAuth 2.0 authentication, Presidio PII redaction, Cloudinary storage, circuit-breaker fault tolerance, and Server-Sent Events (SSE) streaming.

MedMira — AI Prescription Management System | Python, MongoDB, GLiNER, Docker, GCP 2025

- Developed OCR and medical NER pipeline using Google Vision API and GLiNER Bio-Med to extract and structure medication names, dosages, and administration schedules from handwritten and printed prescription images.
- Containerized and deployed backend on Google Cloud Run with GitHub Actions CI/CD; integrated Twilio API for automated WhatsApp medication reminder delivery.

Financial Signal Engineering Pipeline | Python, Docker, Streamlit, MLOps 2025

- Engineered modular MLOps pipeline for financial time-series signal generation, implementing volatility-aware strategies including z-score normalization and hybrid ensemble methods with rolling statistical features.
- Built real-time Streamlit analytics dashboard with Docker containerization, deterministic signal outputs, and comprehensive error handling for production-ready deployment.

Uber Trip Demand Forecasting | Python, XGBoost, Flask, Scikit-learn 2024

- Built time-series demand forecasting system integrating weather and temporal features; achieved 9.52% MAPE and 0.966 R-squared on 250,000+ trip records using a stacked ensemble of XGBoost, Random Forest, and Gradient Boosted Regression Trees.
- Deployed production Flask REST API with 100ms average response latency for real-time ride demand prediction, enabling scalable inference at production scale.

CERTIFICATIONS & ACHIEVEMENTS

- Google Business Intelligence Professional Certificate — Google (2024)
- Machine Learning Operations Fundamentals — Google Cloud Skill Boost (2025)
- AWS Cloud Technical Essentials — Amazon Web Services (2026)
- Portfolio: 3+ production ML systems deployed with live demos and open-source code on GitHub